

Institute of Space Technology	Form No	IST-Dean-F-23/00
FYP Selection Form	Issue Date	Fall 2026
Grading Policy on Final Year Projects	Department	Department of Space Science

Final Year Project (FYP) Defense Proposal Form

University/Institute Detail:

Name of University/Institution:	Institute of Space Technology	Physical Address:	1, Islamabad Highway, Islamabad, 44000
Telephone & Fax No:	+92-51-9075100 +92-51-9273310	Department Name:	Department of Space Science
City:	Islamabad	Province:	Islamabad Capital Territory (ICT)

Nominated Project Details:-

Project Supervisor Name and Designation:	Dr. Munawar Ali Shah Assistant Professor	Contact Details:	munawar.shah@ist.edu.pk
Project Supervisor Qualification:	Ph.D. in Geodesy and GNSS	No. of Publications of Supervisor:	Over 100 peer-reviewed journal and conference publications
Students Name(s):	1. Mashaf Majeed Janjua 2. Haider Taqveen	Students Mobile No:	1. +92-320-5409300 2. +92-325-9514052
Students CGPA:	1. 2.31 2. 2.65	Students Email:	1. 230601017@ist.edu.pk 2. 230601020@ist.edu.pk
Degree Program / Title:	BS Space Science	Area of Specialization:	Remote Sensing & GIS (RS/GIS)

Final Year Project Details:

A. Project Title:	Multi-Agent Deep Reinforcement Learning for UAV Swarm Navigation in a Geospatial Digital Twin
B. Project Start Date:	September 01, 2026 (Fall Semester 2026)
C. Project Finish Date:	June 30, 2027 (Spring Semester 2027)

D. Project Summary (less than 200 words):

Coordinating autonomous Unmanned Aerial Vehicle (UAV) swarms for search and rescue, disaster assessment, and tactical surveillance in dense urban topographies presents severe trajectory and collision challenges. Physical multi-drone testing is prohibitively expensive, safety-critical, and non-repeatable across hazardous disaster events. This project develops an AI-enabled Geospatial Digital Twin that reconstructs simulation-ready, watertight 3D operational environments from sub-meter optical satellite imagery (SpaceNet 2) and normalized Digital Surface Models (nDSM). Operating within this physics-accurate twin, a Multi-Agent Deep Reinforcement Learning (MADRL) architecture—leveraging Multi-Agent PPO (MAPPO) with Centralized Training and Decentralized Execution (CTDE)—is trained to coordinate 6-DOF autonomous UAV swarms. Drone agents utilize simulated 32-beam LiDAR and GPS-degraded kinematic state estimation to learn cooperative area coverage, dynamic obstacle avoidance, catenary powerline traversal, and acoustic beacon localization. Swarm performance is quantitatively benchmarked against classical 3D A* and decentralized RRT* algorithms across mission completion time, energy efficiency, and safety margins. A real-time 3D mission control HUD provides telemetry observation and spatial analytics for mission commanders. [Exact Word Count: 159 words Verified strictly < 200 words]
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E. Project Objectives:

1. Automated 3D Digital Twin Reconstruction: To develop an automated geospatial reconstruction toolchain converting high-resolution optical satellite imagery (SpaceNet 2) and DEM topographic elevations into watertight, collision-ready 3D manifold meshes with localized micro-hazards (catenary powerlines and vegetation volumes). 2. 6-DOF Aerodynamic Swarm Formulation: To formulate and simulate 6-DOF quadrotor kinematic and aerodynamic agent models subject to rotor thrust limits, gyroscopic drag, 32-beam LiDAR emulation, and acoustic rescue beacon chirps. 3. Multi-Agent Reinforcement Learning Engine: To design and train a scalable Multi-Agent Deep Reinforcement Learning (MADRL) framework based on Multi-Agent PPO (MAPPO) with Centralized Training and Decentralized Execution (CTDE) for cooperative swarm navigation. 4. Dynamic Disaster Scenario Emulation: To simulate stochastic, high-entropy environmental conditions, including moving obstacles, catenary wire entanglements, wind shear perturbations, individual motor failures, and intermittent communication loss. 5. Quantitative Benchmarking & Safety Evaluation: To benchmark swarm flight metrics against classical centralized 3D A* and decentralized RRT* baselines across mission completion duration, area coverage rate, collision frequency, and energy consumption. 6. Tactical Mission Control Interface: To construct an interactive 3D WebGL tactical HUD and mission telemetry dashboard for real-time visualization of UAV trajectories, battery states, sensor feeds, and mission progress.

F. Project Implementation Method:

Stage 1 — Satellite Semantic Extraction & Topography (src/data_prep/): The framework consumes 0.3m optical satellite imagery from SpaceNet 2 Las Vegas. A deep convolutional U-Net with ResNet-34 backbone is fine-tuned to extract building footprints (test IoU 0.8062, F1 0.8927). Building masks are vectorized into georeferenced polygons in WGS84 (EPSG:4326). Ground elevations are queried from AWS Terrarium DEM tiles using sub-millisecond bilinear interpolation, while normalized Digital Surface Models (nDSM) and solar shadow trigonometry validate structural building heights. Stage 2 — 3D Digital Twin Mesh Manifolding & Micro-Hazards (src/geometry/): Extracted polygon cadastre footprints are converted into watertight, manifold 3D polygon meshes via automated Blender Python (bpy) extrusion and polygon triangulation. Critical micro-hazards are procedurally modeled into the airspace: catenary powerlines using hyperbolic cosine equations ($y = a \cdot [\cosh(x/a) - 1]$) between transmission poles, and vegetation canopy exclusion zones using SAVI vegetation masks. Stage 3 — Real-Time Physics Simulator & 6-DOF UAV Kinematics (src/physics_sim/): The reconstructed geospatial environment is imported into a real-time simulation engine (Unreal Engine 5 Chaos Physics / lightweight headless 3D simulation). Heterogeneous quadrotor agents are modeled under full 6-DOF rigid-body kinematics (rotor thrust, aerodynamic drag, gravitational torque, and battery energy discharge). Each UAV is instrumented with an emulated 32-beam spherical LiDAR rangefinder and an omnidirectional acoustic receiver detecting emergency chirps.

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F. Project Implementation Method (Continued):

Stage 4 — Multi-Agent Deep Reinforcement Learning Swarm Engine (src/models/): Autonomous swarm coordination is governed by Multi-Agent Proximal Policy Optimization (MAPPO) with Centralized Training and Decentralized Execution (CTDE). During training, a centralized value network observes the global spatial state of the digital twin; during execution, decentralized actor networks generate continuous rotor velocity commands based strictly on local 32-beam LiDAR range vectors and inter-agent relative telemetry. A Pareto multi-objective reward function optimizes mission completion, formation cohesion, and collision avoidance. Stage 5 — Benchmarking & Tactical Mission HUD: The converged MAPPO swarm policy is benchmarked against centralized 3D A* and decentralized RRT* algorithms under dynamic scenarios (moving obstacles, catenary powerlines, motor dropouts, and communication packet loss). An interactive WebGL 3D Mission Control HUD renders real-time UAV trajectories, telemetry cards, and quantitative safety reports.

G. Key Milestones of the Project with Dates:

S.No.	Elapsed Time & Dates	Milestone	Key Deliverables
1.	Month 1 (Sep 2026)	Literature Review, System Requirements & Mathematical Modeling	Comprehensive literature review, 6-DOF kinematic formulations, and system architecture specification document.
2.	Months 2–3 (Oct–Nov 2026)	Satellite Extraction Pipeline & 3D Geospatial Digital Twin	Trained U-Net footprint segmentation model, vectorized building cadastre, nDSM height model, and automated watertight 3D mesh generator.
3.	Months 4–5 (Dec 2026–Jan 2027)	Physics-Based Multi-UAV Simulation & MAPPO RL Training	Multi-agent simulation engine with 32-beam LiDAR emulation, converged MAPPO policy weights, and acoustic beacon search behavior.
4.	Months 6–7 (Feb–Mar 2027)	Dynamic Scenario Stress-Testing & Algorithmic Benchmarking	Empirical test results across moving obstacles, wire entanglements, and UAV failures; quantitative comparisons vs 3D A* and RRT*.
5.	Month 8 (Apr–May 2027)	Tactical HUD Integration, Final Demonstration & Thesis Defense	Interactive 3D WebGL mission control interface, complete software repository, finalized BS Space Science thesis dissertation, and oral defense.

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H. Final Deliverable of the Project:

- 1. End-to-End Software System:** Fully automated software toolchain for satellite building extraction, 3D mesh manifold reconstruction, and multi-agent flight simulation.

2. Autonomous Swarm Policy Checkpoints: Converged PyTorch neural network checkpoints implementing decentralized cooperative navigation and hazard avoidance.

3. Quantitative Simulation & Comparative Study: Comprehensive comparative benchmarking dataset validating swarm flight performance against 3D A* and decentralized RRT*.

4. High-Fidelity Simulator & Tactical HUD: 3D simulation environment and interactive WebGL Mission Control HUD for real-time mission telemetry and spatial analytics.

I. Please Specify Technical Details of Final Deliverable:

- 1. Virtual Geospatial Environment:** Cadastre-grade 3D environment generated from SpaceNet 2 satellite tiles, georeferenced in WGS84 (EPSG:4326), featuring accurate building geometries, terrain topography, and restricted flight volumes.

2. Digital Twin State Engine: Real-time spatial state manager synchronizing environmental changes, multi-agent position histories, and dynamic threat boundaries.

3. Multi-UAV Simulation Module: Physics-based kinematic engine simulating heterogeneous quadrotor agents under 6-DOF rigid-body dynamics, aerodynamic rotor drag, and battery discharge curves.

4. AI Swarm-Control Module: MAPPO multi-agent reinforcement learning architecture utilizing Centralized Training with Decentralized Execution (CTDE) for collision-free cooperative navigation.

5. Dynamic Scenario Generator: Simulation module injecting moving obstacles, localized wind gusts, catenary wire obstacles, and UAV communication packet dropouts.

6. Mission Management Module: Mission coordinator managing search patterns, acoustic rescue beacon homing, formation tracking, and automated return-to-base protocols.

7. Performance Evaluation Module: Automated logging engine tracking mission completion duration, area coverage rate, collision frequency, and energy consumption (Wh).

8. Tactical Visualization Interface: Interactive 3D WebGL HUD rendering swarm trajectories, drone health telemetries, and operational status cycling.

9. Experimental Benchmark Suite: Standardized evaluation framework executing head-to-head performance comparisons against centralized 3D A* and decentralized RRT* algorithms.

J. Equipment required for making prototype/working model:

- 1. Development Workstation:** Standard research workstation or laptop equipped with a multi-core processor (Intel Core i7/i9 or AMD Ryzen) and 16+ GB RAM.

2. GPU Compute Capability: Dedicated NVIDIA GPU (GeForce RTX/GTX series, CUDA 12.x compatible) for accelerated deep neural network training and real-time 3D simulation physics.

3. GIS & Simulation Software Toolchain: Python 3.10+ virtual environment, PyTorch deep learning framework, GDAL/Rasterio/GeoPandas geospatial libraries, Blender 4.x (bpy), and Unreal Engine 5.4.

4. Earth Observation Datasets & Connectivity: High-speed internet connectivity for streaming AWS Terrarium DEM tiles and SpaceNet 2 satellite imagery.

5. Hardware-in-the-Loop Demonstration (Optional): An optional physical quadrotor or Pixhawk flight controller telemetry link may be incorporated for ground station telemetry playback if laboratory hardware is made available.

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K. Benefits of the Project:

- 1. Risk-Free, Low-Cost Swarm Development:** Enables rapid prototyping and verification of multi-UAV swarm control policies without incurring drone crash costs or flight-safety hazards.
- 2. Scientific Repeatability:** Provides a deterministic, repeatable operational environment for rigorous scientific comparison across disparate multi-agent algorithms under identical environmental states.
- 3. Robustness in High-Entropy & GPS-Degraded Environments:** Enables evaluation of swarm adaptability under communication outages, moving obstacles, micro-hazards (catenary powerlines), and individual agent motor failures.
- 4. Direct Fusion of Space Science & Autonomous Robotics:** Demonstrates seamless integration of satellite remote sensing, photogrammetric DEMs, and GIS spatial analytics directly into autonomous robotic navigation pipelines.
- 5. Search and Rescue (SAR) & Disaster Assessment:** Directly applicable to post-disaster reconnaissance (earthquake structural collapse, flood assessment, acoustic survivor locator homing).
- 6. Quantitative Multi-Objective Benchmarking:** Establishes empirical baselines comparing cooperative reinforcement learning against classical centralized 3D A* and decentralized RRT* algorithms.
- 7. Foundation for Physical Hardware-in-the-Loop (HIL):** Establishes standard MAVLink telemetry and control interfaces readily transferable to physical multi-UAV platforms in future research.

It is certified that the FYP titled “Multi-Agent Deep Reinforcement Learning for UAV Swarm Navigation in a Geospatial Digital Twin” has been approved and is being undertaken by the above-mentioned students (Mashaf Majeed Janjua, Reg. No. 230601017, and Haider Taqveen, Reg. No. 230601020) as their Final Year Project.

It is undertaken that the undersigned have understood the “terms & conditions” of the program, and further reiterate that, if the subject FYP is approved for funding, the disbursed funds shall be utilized as per “terms & conditions” and the undersigned will be liable to reimburse/refund the unutilized amount and other cost not approved by the IST, if any.

It is further undertaken that after utilization of funds, the said “Fund Utilization Report” shall be furnished along with the other required deliverables as and when required by the IST.

1. Name, Designation & Signature of Supervisor: Dr. Munawar Ali Shah Assistant Professor, Department of Space Science Institute of Space Technology, Islamabad Signature: _____ Date: _____	2. Name & Signatures of HoD: Head of Department Department of Space Science Institute of Space Technology, Islamabad Signature: _____ Date: _____
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